

Smart Waste Management System For Metropolitan Cities

Final Year Project Documentation

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Project Domain: IoT, Smart Cities, Web Development, AI

1. Project Overview

The **Smart Waste Management System (SWMS)** is an advanced full-stack web application that simulates a real-world IoT-powered waste monitoring platform for metropolitan cities. It integrates real-time bin fill-level tracking, AI-based route optimization, live analytics dashboards, and a REST API for IoT device connectivity.

The system monitors 12 smart bins distributed across 6 zones of the **Pune Metropolitan Region**, providing municipal authorities with actionable insights to optimize collection schedules, reduce fuel costs, and maintain city hygiene standards. Built using Python Flask, Leaflet.js maps, Chart.js analytics, and a professional dark-sidebar UI, this project demonstrates a complete end-to-end smart city solution aligned with UN Sustainable Development Goals 11, 12, 14, and 15.

2. Objectives

- **Design and develop** a real-time IoT bin monitoring system using simulated ultrasonic sensor data.
- **Implement** an AI-powered route optimization algorithm using Greedy Nearest-Neighbour logic with Haversine distance calculations.
- **Build** a professional analytics dashboard featuring live charts, KPI cards, and zone-wise analysis.
- **Create** an interactive city map with color-coded bin markers and animated truck route visualizations.
- **Develop** a complete REST API for IoT device integration and mobile app connectivity.
- **Align** the core solution with UN SDG 11 (Sustainable Cities) and SDG 12 (Responsible Consumption).

3. Technologies Used

Layer	Technology	Purpose
Backend	Python 3 + Flask 3.1	Web server, routing, and REST API development
Frontend	HTML5 + Custom CSS3	Professional dark-sidebar user interface
Maps	Leaflet.js + OpenStreetMap	Interactive city map tracking bin markers
Charts	Chart.js 4.x	Dynamic bar, doughnut, pie, and line charts
Algorithm	Greedy Nearest-Neighbour	Optimal collection truck route planning
Distance	Haversine Formula	GPS-accurate distance calculations
IoT Sim	Time-seeded RNG	Realistic fill-level behavior simulation
API	Flask REST (JSON)	IoT device and mobile application integration
Version Control	Git & GitHub	Source code management

4. Folder Structure

Plaintext

```
Smart-Waste-Management-System/
├── Documentation/
│   └── Project Documentation.docx
├── static/
│   └── css/
│       └── style.css
├── templates/
│   ├── sidebar.html           # Reusable sidebar component
│   ├── home.html             # Landing page
│   ├── dashboard.html        # Live analytics dashboard
│   ├── map.html              # Interactive map route
│   ├── bins.html             # Bin monitoring grid
│   └── reports.html           # Weekly KPI reports
└── app.py                     # Main Flask application
```

requirements.txt

README.md

(Structure adapted from)

5. System Architecture

5.1 Overall Flow

- IoT sensors (simulated) report bin fill levels every 15 minutes.
- The Flask backend processes the incoming data, computes the optimal route, and generates alerts.
- Jinja2 templates render the live data cleanly to the web browser.
- Client-side JavaScript (Chart.js, Leaflet.js) renders the interactive visual assets.
- REST API endpoints allow external IoT devices and mobile apps to push or pull data.

5.2 Bin Network Deployment

The system actively coordinates 12 Smart Bins across 6 designated zones:

Bin ID	Location	Zone	Waste Type	Capacity
BIN001	MG Road Junction	Zone A	General	100 L
BIN002	FC Road Market	Zone A	Recyclable	100 L
BIN003	Shivaji Nagar Station	Zone B	Organic	150 L
BIN004	Koregaon Park Gate	Zone B	General	100 L
BIN005	Baner Tech Hub	Zone C	Recyclable	200 L
BIN006	Hinjewadi IT Park	Zone C	General	200 L
BIN007	Wakad Market	Zone C	Organic	100 L
BIN008	Kothrud Square	Zone D	General	150 L
BIN009	Sinhagad Road	Zone D	Recyclable	100 L
BIN010	Hadapsar Industrial	Zone E	Hazardous	200 L
BIN011	Viman Nagar Airport Rd	Zone E	General	100 L

Bin ID	Location	Zone	Waste Type	Capacity
BIN012	Aundh Central Park	Zone F	Organic	150 L

6. Key Features

6.1 Live Dashboard

- **4 KPI Stat Cards:** At-a-glance metrics for Total Bins, Critical Bins ($\geq 85\%$), Warning Bins (60-84%), and Average City Fill Percentage.
- **Chart.js Grouped Bar Chart:** Compares weekly waste collected vs. recycled (in kg).
- **Doughnut Chart:** Visualizes current bin status distribution (OK / Warning / Critical).
- **Real-time Alert Feed:** Shows urgent updates with severity icons and timestamps.
- **Zone-wise Grid:** Tracks the average fill percentage across all 6 city zones.
- **Actionable Critical Table:** Lists high-priority bins with an immediate collection action button.

6.2 Interactive Map

- Integrated Leaflet.js map centered precisely on the **Pune Metropolitan Region**.
- **Color-Coded Status Markers:** Red (Critical), Yellow (Warning), and Green (OK).
- **Dynamic Pop-up Cards:** Clickable components revealing fill levels, waste type, and predicted time until full.
- **AI Route Polyline:** Displays an animated dashed polyline tracking the optimized truck route.
- **Numbered Stops:** Visually labels route milestones sequential order (1, 2, 3\dots).
- **Route Statistics:** Live tracker for total stop counts, total distance (km), and estimated travel time.

6.3 Bin Monitor Grid

- A fully responsive card layout showing all 12 individual bins equipped with precise fill bars and status badges.
- Advanced filtering systems categorized by **Zone (A-F)** and **Waste Type** (General, Recyclable, Organic, Hazardous).
- Heuristic-driven predictive countdown estimating time remaining until bins reach capacity.
- Color-coded top borders corresponding directly to live bin status severity.

6.4 Reports & KPIs

- **Analytical Visuals:** Bar charts mapping weekly collection trends and pie charts detailing waste type distributions.
- **Historical Tracking:** A 7-day line chart mapping city-wide fill levels.

- **Aggregated Metrics:** Total waste collected, total volume recycled, overall recycling rate percentage, and active bin numbers.

7. Route Optimization Algorithm

7.1 Greedy Nearest-Neighbour

The core software leverages a **Greedy Nearest-Neighbour** routing algorithm to plot the collection vehicle's path. To optimize municipal fuel resources and minimize traffic footprint, **only bins with Critical status ($\geq 85\%$ full) are injected into the active pathing array.**

7.2 Haversine Distance Formula

To compute distance matrices across live GPS coordinates, the system utilizes the mathematical **Haversine formula**, accounting elegantly for the Earth's spherical curvature:

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta \text{lat}}{2} \right) + \cos(\text{lat}_1) \cos(\text{lat}_2) \sin^2 \left(\frac{\Delta \text{lng}}{2} \right)} \right)$$

The computed distance output maps accurately in kilometers, which allows the application to project highly realistic travel times based on a standard fleet vehicle averaging a city speed of 30 km/h .

8. IoT Sensor Simulation

Each physical bin unit relies on a specialized time-seeded random number generator designed to replicate genuine ultrasonic sensor behaviors:

- **Hourly Consistency:** Generated levels remain uniform within any specific hour block, mirroring real hardware capturing samples every 15 minutes.
- **Accumulation Trends:** Generated values scale smoothly across sequential hour variations to map true-to-life urban garbage accumulation.
- **Unique Seeding:** Every device maintains a unique seed tied explicitly to its unique `Bin ID`, ensuring unique data feeds across different locations.
- **Threshold Triggers:**
 - **OK:** $< 60\%$
 - **Warning:** $60\% - 84\%$
 - **Critical:** $\geq 85\%$

9. REST API Reference

Endpoint	Method	Description	Response Format
/api/bins	GET	Fetches all current bin statuses and fill metrics	JSON Array
/api/bins/<id>	GET	Fetches real-time localized data for a single unit	JSON Object
/api/bins/<id>/update	POST	Entry point for physical IoT hardware to push data	Action Decision
/api/route	GET	Pulls the AI-optimized collection path array	Ordered JSON List
/api/stats	GET	Provides city-level aggregated data summaries	Summary JSON

9.1 Sample API Response (/api/stats)

JSON

```
{
  "total_bins": 12,
  "critical": 4,
  "warning": 3,
  "ok": 5,
  "avg_fill": 61.5
}
```

(Adapted from)

10. Setup & Run Instructions

To deploy and execute the platform locally, follow these steps:

Step 1: Install Dependencies

Ensure you have Python installed, then run the package manager via terminal:

```
Bash
pip install -r requirements.txt
```

(Adapted from)

Step 2: Launch the Flask Application

Run the primary execution script:

```
Bash
python app.py
```

(Adapted from)

Step 3: Access the Interface

Open your web browser and navigate to the following local address:

URL: `http://localhost:5000`

11. UN SDG Alignment

SDG Goal	Title	Core Project Contribution
SDG 11	Sustainable Cities	Real-time smart bin analytics prevent overflow, keeping urban pathways cleaner.
SDG 12	Responsible Consumption	Integrated recycling rate metrics foster better waste classification and tracking.
SDG 14	Life Below Water	Proactive collection mitigates loose debris runoff from entering local waterways.
SDG 15	Life on Land	Optimized management drastically reduces plastic scatter and general land pollution.

12. Conclusion

The **Smart Waste Management System** successfully highlights an industry-standard implementation of modern web technology paired with intelligent routing algorithms to tackle modern urban planning crises. By fusing real-time monitoring maps, an automated route system, professional metric graphs, and accessible REST endpoints, it functions as a highly valuable tool for municipal city developers.

The codebase is engineered to scale easily. Future iterations can cleanly substitute the time-seeded simulators for live physical microcontrollers (e.g., Arduino or Raspberry Pi utilizing MQTT architecture), deploy immediate email/SMS notifications, or build Machine Learning models for predictive waste forecasting.

Project Links

- **GitHub Repository:** [Smart-Waste-Management-System-Main](#)

- **Video Demonstration:** [Google Drive Demo Link](#)